

Safe reprocessing of used surgical instruments



A potential approach for safe and efficient ultrasonic cleaning in Bandelin's SONOREX ultrasonic bath.

The reprocessing of medical instruments with ultrasound enables a fast instrument cycle and gentle intensive cleaning by disinfection and cleaning. The usual damage and wear of the instruments by manual "brushing" is thus excluded. In the dental field, SONOREX ultrasonic baths are used for instrument preparation and impression mould cleaning. Due to the insert cups specially developed for this application, impression mould cleaning and instrument disinfection with different cleaning preparations can be carried out simultaneously in one unit.

The SONOREX ultrasonic baths use the effect of cavitation. They contain piezoelectric transducers under their oscillating tank bottoms, which cause mechanical oscillations. As a result, microscopically small bubbles are continuously formed in the bath liquid, which release energy upon imploding and generate microcurrents. During the cleaning process, it causes contamination to 'blast off' from the surfaces and hard-to-reach places.

Which instruments can be reprocessed with ultrasound?

- General instruments such as scissors, needle holders, tweezers, forceps, trocars, scalpels.
- Micro-instruments from neurosurgery and ophthalmology.
- MIS instruments such as dismountable tubular shaft instruments, microclamps, etc.
- Endoscope accessories such as biopsy forceps, slings, valves.
- ECG, EEG electrodes.

Advantages of instrument reprocessing in an ultrasonic bath

- Fast instrument circulation.
- Time reduction of the chemical disinfection to 5 minutes.
- Protection of the instruments.
- Short contact of the instruments with disinfection and cleaning liquids.
- No risk of corrosion.
- Economical use of water, chemicals and energy.
- High cleaning effect in hard-to-reach places, such as boreholes, joints or grooves, without mechanical damage.

Application showcase

Which ultrasonic bath do I choose?



- Size of the cleaning objects determines the tub size and thus the type of appliance.
- Dimensions of the hanging baskets must be observed.
- To avoid overloading the unit, it is advisable to choose the next larger unit. This also gives scope for further applications.

What to consider when using the ultrasonic bath

- The items to be cleaned must not lie directly on the bottom of the tub.
- Do not stack instruments and do not overload baskets.
- Open articulated instruments, e.g. forceps or scissors, completely and disassemble if necessary.

Is heating required?

Unit without heating

- For simultaneous disinfection and cleaning after dry filing.
- Disinfectant liquids must not be warmed up.
- At temperatures above 40°C, there is a risk of protein coagulation.

Unit with heating:

- For cleaning after wet filing or for basic cleaning.
- Impurities are removed more quickly.

- Instruments must be completely covered with solution.
- Air from cavities must escape recommended liquids.
- STAMMOPUR concentrates were specially developed for disinfection and cleaning in the ultrasonic bath.
- The corresponding microbiological reports are available for the disinfection time reduction in the ultrasonic bath.
- Flammable liquids such as alcohols and aggressive cleaning liquids such as acids and salt solutions are not permitted for use.
- Disinfection or cleaning is not possible with water alone, that is, without suitable additives.

Operation process-Bandelin ultrasonic baths

